Unit Weight

SKU:U030













Description

Unit Weight is a weighing unit. It integrates the 24-bit A/D converter chip **HX711**, which is specifically designed for high-precision electronic scales. This design utilizes only **Channel A**, amplifying the acquired signal through its built-in low-noise programmable gain amplifier (PGA). The programmable gain of Channel A can be set to 128 or 64, corresponding to full-scale differential input signal amplitudes of ±20mV or ±40mV, respectively. All control signals are driven by pins, eliminating the need to program the chip's internal registers. This product is suitable for high-precision weighing systems, industrial automation control, logistics and warehouse management, food processing, pharmaceutical measurement, and other fields.

Features

- HX711:
  - High-precision 24-bit ADC
  - Programmable gain amplification: 32, 64, and 128
  - 10SPS output data rate
- Development Platforms:
  - UiFlow1
  - UiFlow2
  - Arduino IDE
  - ESP-IDF

- PlatformIO

Includes

- 1 x Unit Weight
- 1 x HY2.0-4P Grove cable (20cm)
- 1 x HT3.96-4P Terminal

Applications

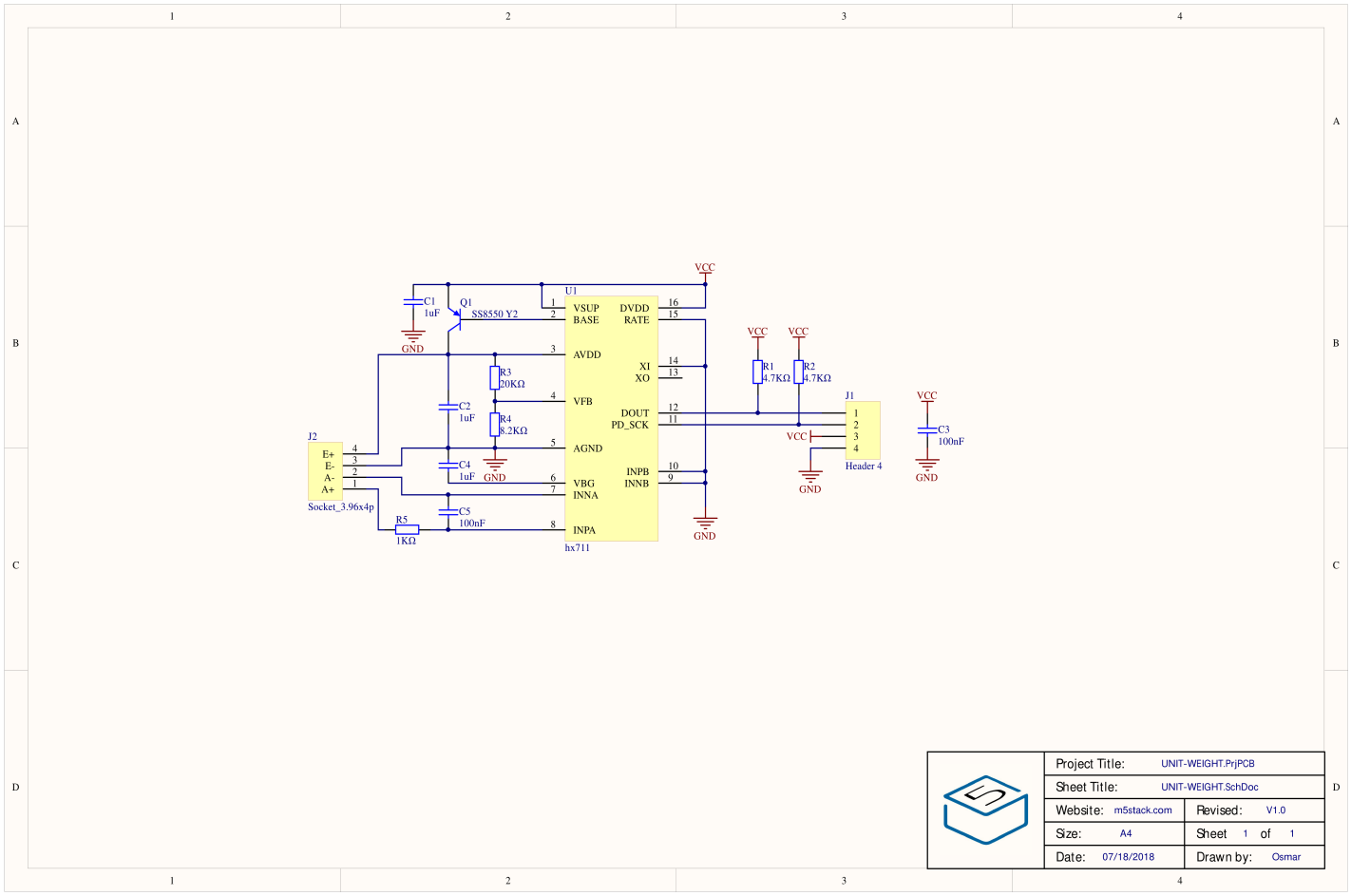
- Miniature weight scales
- Kitchen scales

Specifications

Specification	Parameters
Product Size	32.0 x 24.0 x 10.2mm
Product Weight	5.1g
Package Size	138.0 x 93.0 x 11.2mm
Gross Weight	13.0g

Schematics

- Unit Weight Schematics PDF

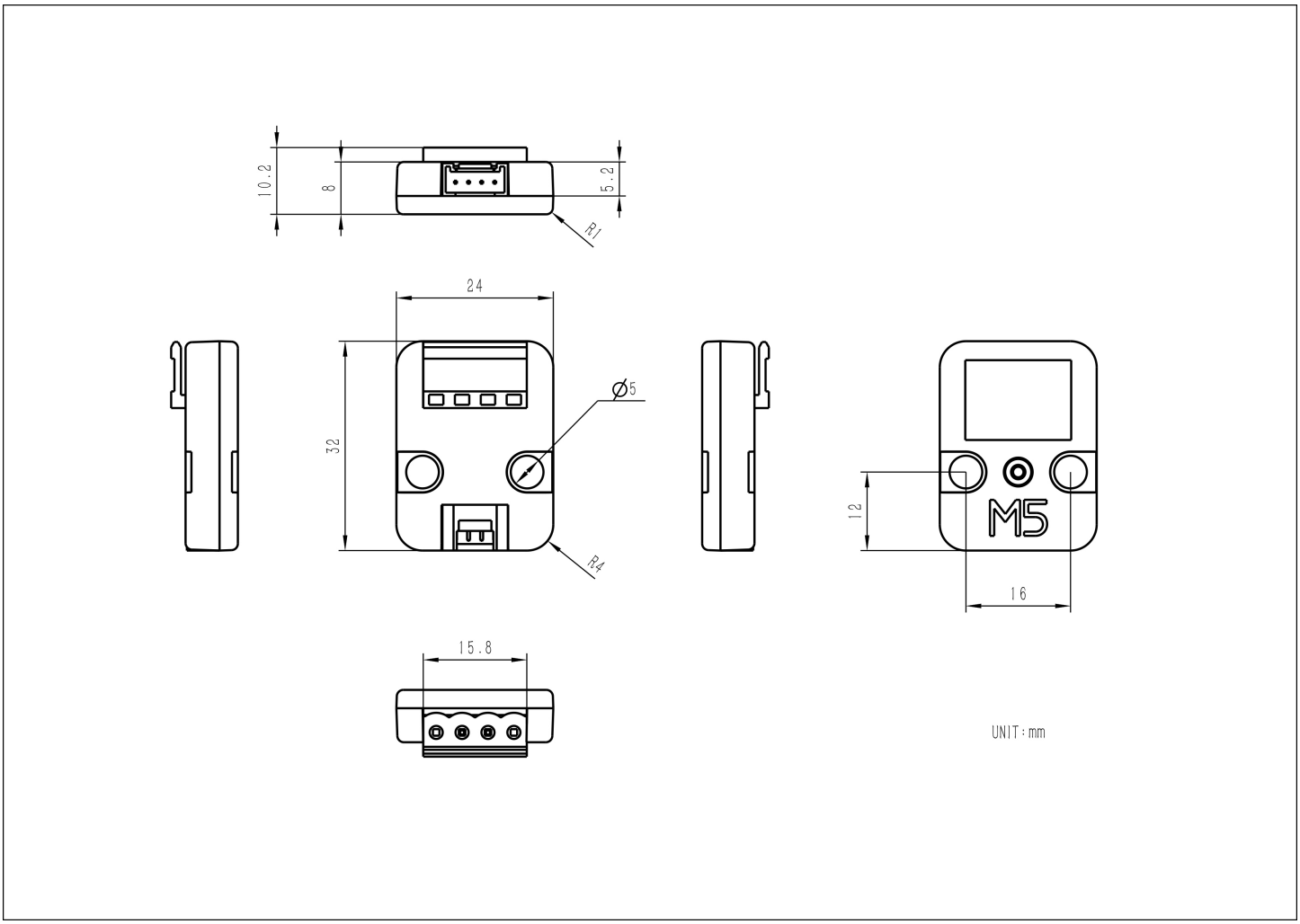


PinMap

Unit Weight

HY2.0-4P	Black	Red	Yellow	White
PORT.B	GND	5V	CLK	DAT

Model Size



Structure

- Unit Weight Structure Files

Datasheets

- HX711

Softwares

Arduino

- Unit Weight Example with M5Core

UiFlow1

- Unit Weight UiFlow1 Docs

UiFlow2

- Unit Weight UiFlow2 Docs

EasyLoader

Easyloader	Download Link	Notes
Unit Weight Example with M5Core	download	/

Video

Product Comparison

Product Compare					
	Unit Scales	**Scales Kit**	**Unit Mini Scales**	**Unit Weight**	**Unit Weight-I2C**
Communication Protocol	I2C	ADC	I2C	ADC	I2C
Chip Solution	STM32+HX711	HX711	STM32+HX711	HX711	STM32+HX711
Sensor Range	0 ~ 20kg	0 ~ 200kg	0 ~ 5kg	Depends on the strain gauge and controller chip	Depends on the strain gauge and controller chip